BEGINNER GUIDE

Scroll-video websites: why, when, and how

Why a scrubbed video feels silkier than a pile of images, when to use which — and a step-by-step plan to make Aisle's whole-page shopping film with ChatGPT and Claude.

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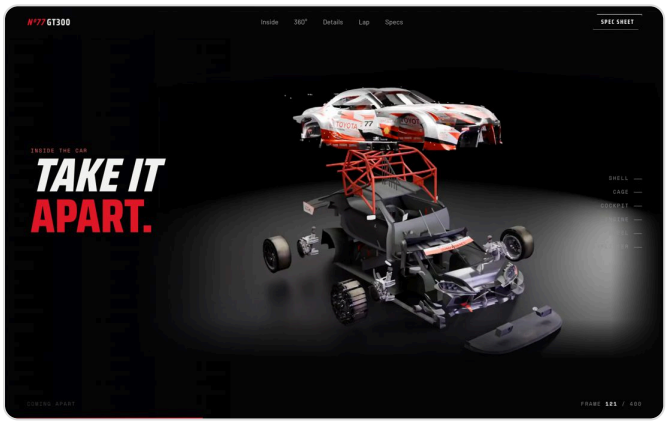
Written for Amartya Ghosh

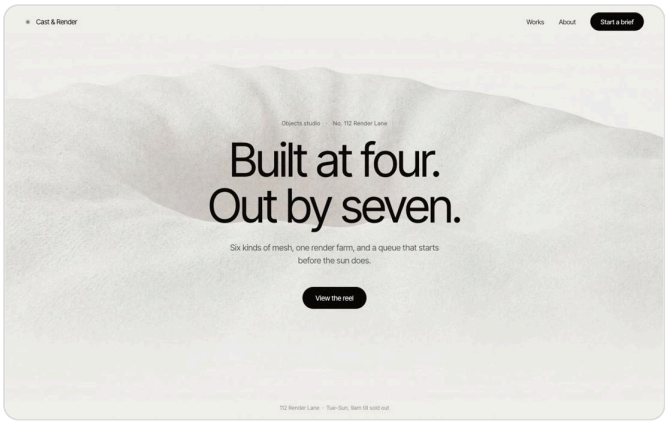
2 Nov 2026 · Examples: Cast & Render · N°77 GT300 · Aisle

01 Part 1 — The big idea: scroll is a remote control

Every “cinematic” scroll site is a **film you control with the scrollbar**. Scroll down = play forward, scroll up = rewind, stop = pause. The film can be stored two ways:

	A pile of images	One video file
Analogy	a **flip-book**: 400 separate cards, you flip to the right one	a **DVD**: one disc, you jump to a moment with the remote
What the browser does	loads 400 files, draws the right one on a canvas	loads 1 file, its video chip shows the right moment
Our example	N°77 car, first version (400 WebP images, 20 MB)	Cast & Render; N°77 car now (1 MP4, 14.6 MB)





Left: the N°77 car story (now a video). Right: Cast & Render (a video from the start).

02 Why video feels silkier — three reasons

1. A built-in, specialised chip. Phones and laptops have hardware made only for showing video (the same chip that plays YouTube). Drawing big images on a canvas uses the general graphics path instead. *Analogy: a professional projector vs. holding photos up one by one.*

2. It glides instead of jumping (“eased seeking”). Your scroll says “be at 5.0 s”. The page doesn't jump there; every screen refresh it moves **11.5% of the remaining distance**: 4.0 → 4.12 → 4.22 → 4.31 … arriving softly. *Analogy: a lift that slows down before the floor instead of stopping dead.* Tiny scrolls make tiny moves, so slow scrolling is smooth.

3. It's all in memory before you start. The whole file is downloaded first (the loader shows the real %), so no jump ever waits for the internet. *Analogy: copying the film to your laptop before the flight instead of streaming on plane Wi-Fi.*

> Same number of pictures (our car is 400 frames either way) — the video is smoother because of HOW they're shown, not how many.

03 The one rule that makes it work: “all-intra”

Normal videos save space by storing a **full picture only every few seconds** and, in between, just **the differences** — like a “spot the difference” note: *“same as before, but the ring moved 2 mm”*. To show a moment in the middle, the player must rebuild from the last full picture. Fine when playing forward, **terrible when you scroll backwards or jump**.

An **all-intra** video stores **every frame as a full picture**. Any moment shows instantly, forwards or backwards. The file is bigger, but scrubbing becomes perfect.

```
# Is my video all-intra? (I = full picture)
ffprobe -v error -select_streams v -show_frames -show_entries frame=pict_type -of csv=p=0
video.mp4 | sort | uniq -c

# Make any video all-intra for scrolling
ffmpeg -i input.mp4 -c:v libx264 -preset slow -crf 22 -pix_fmt yuv420p \
       -x264-params keyint=1:scenecut=0 -movflags +faststart -an scrub.mp4

# Phone version (smaller)
ffmpeg -i input.mp4 -vf scale=960:-2 -c:v libx264 -preset slow -crf 24 -pix_fmt yuv420p \
       -x264-params keyint=1:scenecut=0 -movflags +faststart -an scrub-m.mp4
```

04 When to use video — and when images win

Use VIDEO when…	Use IMAGES when…
The scroll plays **one story along one line** (forward / back): a product reveal, a car taking itself apart, a dish being made	You need **free, instant control**: a 360° turn that follows the cursor and goes round past 360° → 0°
You want **smooth slow scrolling** and a **smaller download**	You need **transparency** (a cut-out product over changing backgrounds) — web video can't be see-through
The film is **short** (≈ 6–15 s ≈ 10–20 MB at 1080p)	The sequence is **very long** (all-intra gets ≈ 1.5 MB per second at 1080p)
You can handle the **iPhone quirk** (play/pause once on first touch — already built in our code)	You want it to work **identically everywhere** with zero tricks

Why the 360° turn can't be a video (the circle problem): in a video, 0° is at the start and 360° at the end. Turning past 360° means jumping from the end back to the start — with gliding, the car would rewind through the whole film and spin backwards a full circle. And a twitchy cursor asks for many jumps per second, which video can't keep up with. Images just draw instantly, wherever you point.

> Rule of thumb: a story → video. A toy you play with → images (or live 3D).

05 How our code does it (the pieces)

```
// 1. Download the whole film, show real %, keep it in memory
fetch("/story.mp4") → read chunks → new Blob(...) → video.src = URL.createObjectURL(blob)

// 2. Every screen refresh: glide toward the scroll position
seekTo = scrollProgress * video.duration;
seekAt += (seekTo - seekAt) * 0.115;
if (!video.seeking) video.currentTime = seekAt;

// 3. Text is tied to the film's frame number
frame = Math.round(seekAt * 40);            // 40 fps
card.classList.toggle("on", frame >= from && frame <= to);

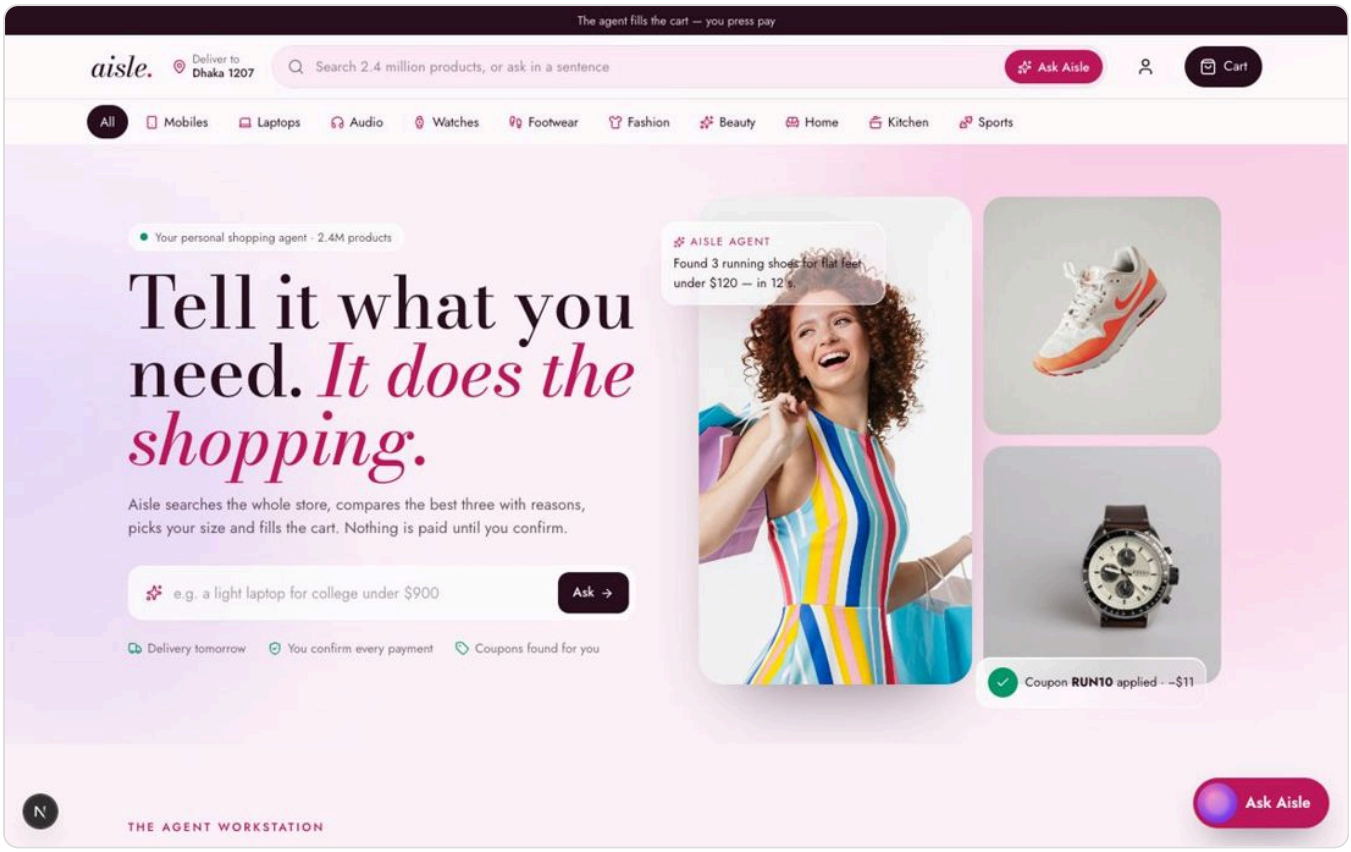
// 4. iPhone: play + pause once on the first touch so it will paint frames
```

Files: `aether/src/scrub.js` (the video engine) and `cast-render/index.html` (the single-file version). Both are explained line by line in their guides.

06 Part 2 — Aisle's whole-page shopping film: the plan

Goal: one **10–12 second film** that plays across Aisle's whole scroll and tells the agent story — *you ask → it searches everything → it compares the best → it fills the bag → you confirm*. Made from AI images and AI video, in Aisle's look: **blush pink, berry, emerald, soft glass, premium studio light**.

Scene	Seconds	What happens	Text on the page
1 · Ask	0–2.5	A frosted-glass shopping bag floats in a blush studio; it opens	“Tell it what you need.”
2 · Search	2.5–5	Dozens of products burst out and orbit in a slow spiral (the “ingredient ballet”)	“It searches 2.4 million products.”
3 · Compare	5–8	Most fade away; three sneakers glide forward into a row and turn	“…and compares the best three.”
4 · Confirm	8–11	The chosen sneaker flies into the bag; an emerald ribbon ties it	“You confirm. It's on its way.”



Aisle today — the film would play behind the hero and the agent workstation.

07 Which tools do what (honestly)

Tool	Can it make pictures / video?	Use it for
ChatGPT Plus — images	✅ images (GPT image model)	The **keyframes**: one perfect still for the start and end of each scene, all in the same style
ChatGPT Plus — **Sora**	✅ short videos (limits depend on your plan — check sora.com)	Turning keyframes into motion (image-to-video)
Claude Pro (chat)	❌ no images or video	Writing and refining prompts, the storyboard, checking consistency, scene timing
Claude Code (me, here)	❌ no generation	Joining clips, making them all-intra, building them into the Aisle site, timing the text
Kling (free credits)	✅ video, very good at products	Backup if Sora's limits run out; supports start + end frame

> Plans and limits change often — if Sora or a feature isn't available on your account, Kling's free daily credits do the same job.

08 Step 1 — Make the keyframes in ChatGPT (images)

Make **all stills first**, in one ChatGPT chat, so the style stays identical. Ask for **16:9, 1920×1080 or larger**. Generate 2–4 versions of each and keep the best. Save them as `k1.png … k5.png`.

```
STYLE (paste at the end of every image prompt):
"Premium e-commerce product photography, seamless blush-pink studio (#FDF2F8), soft diffused top light,
gentle pink reflections, frosted glass and berry-red (#BE185D) accents, ultra realistic, sharp focus,
16:9, empty space on the left third for text, no text, no logos, no people, no hands."

k1 — the closed bag:
"A floating frosted-glass shopping bag with a berry-red ribbon handle, hovering in the centre-right. + STYLE"

k2 — products orbiting:
"The same frosted-glass bag, open, with premium products floating around it in a loose spiral — white sneakers, a silver chronograph watch, black over-ear headphones, a glass perfume bottle, sunglasses, a smartphone, a skincare bottle. + STYLE"

k3 — the best three:
"Three identical premium white running sneakers lined up side by side, floating, side view,
the frosted-glass bag softly out of focus behind them. + STYLE"

k4 — into the bag:
"One white running sneaker gliding into the open frosted-glass bag. + STYLE"

k5 — confirmed:
"The frosted-glass bag closed and tied with a deep emerald-green (#059669) silk ribbon bow,
a soft glow around it. + STYLE"
```

- Keep the bag **in the same place and size** in every still (say “same position as before” and upload the previous image as reference).
- Ask Claude Pro: *“Here are my 5 keyframes — check they match in lighting, colour, angle and bag position, and rewrite any prompt that drifts.”*

09 Step 2 — Animate them in Sora (or Kling)

Make **one short clip per scene**, each going **from one keyframe to the next** (start image → end image). That's what makes the scenes join seamlessly.

Clip	Start → end	Motion prompt
1	k1 → k2	“The bag slowly opens and products float out upward in slow motion, spreading into a gentle spiral around it. Camera stays still.”
2	k2 → k3	“The products slowly orbit, then most drift away out of focus while three white sneakers glide forward into a neat row. Camera slowly pushes in.”

Clip	Start → end	Motion prompt
3	k3 → k4	“The middle sneaker turns and glides into the open bag; the other two fade away. Camera stays still.”
4	k4 → k5	“The bag closes and an emerald silk ribbon wraps around it and ties a bow. Camera stays still.”

```
Settings for every clip:
  • 16:9 · highest resolution your plan allows (1080p) · 3–4 seconds each
  • Add to every motion prompt: "slow, smooth, continuous, cinematic, soft studio light,
    no new objects, products keep their exact shape, no text"
  • If the tool has 'negative prompt': "shaky camera, morphing, melting, extra objects, flicker,
text"
  • Make 3 versions of each clip; keep the one where products don't melt or change
```

> Golden rule: ONE motion per clip, slow, and the camera still. Camera + object moving at once is what makes AI video wobble.

10 Step 3 — Give me the clips; I build it

```
Put the files here:  ~/Downloads/assets/aisle/ai/clip1.mp4 … clip4.mp4

What I'll do:
# 1. join the clips into one film (no gaps)
ffmpeg -f concat -safe 0 -i list.txt -c copy story_raw.mp4
# 2. make it all-intra for smooth scrolling (+ a phone version)
ffmpeg -i story_raw.mp4 -c:v libx264 -crf 22 -pix_fmt yuv420p -x264-params keyint=1:scenecut=0 \
       -movflags +faststart -an public/aisle-story.mp4
# 3. build it into Aisle with the same scrub engine as the car (src/scrub.js):
#    fixed behind the page, text cards timed to scenes, agent workstation over the top
```

Scroll position	Film time	On the page
0–20%	Scene 1	Hero: “Tell it what you need.” + the ask box
20–45%	Scene 2	“It searches 2.4 million products.” + live counter
45–75%	Scene 3	“Compares the best three.” + the comparison cards
75–100%	Scene 4	“You confirm.” + Confirm & pay button

- Before you start, check the image and video tools' terms allow you to use the results on a public site (most do for paid plans).
- Keep the whole film under ~15 MB (≈ 10–12 s at 1080p all-intra).

11 Cheat sheet

Question	Answer
Why is video smoother?	Video chip + gliding (eased seeking) + whole file in memory
What's all-intra?	Every frame a full picture → instant jumps, even backwards
When images?	Cursor-controlled 360°, transparency, very long sequences, zero-trick compatibility
Best length / size?	6–15 s, ≤ 15 MB at 1080p
AI video tip	Keyframes first (same style), start → end frame per clip, one slow motion, camera still
Who does what	ChatGPT: images + Sora video · Claude Pro: prompts & checks · Claude Code: build it